Witle
Academy

SQL PROJECT

Nova Retail Group Database Analysis

Business Intelligence & Data Analytics

Student Name: ______________________________

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Date: _______________________________________

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1. Project Overview

Company Background

Nova Retail Group is a growing retail company operating across multiple regions in South Africa. The company sells consumer products through both physical stores and an online platform. Management needs your help analyzing their data to make better business decisions.

Your Role

You are a Junior Data Analyst tasked with analyzing sales data, customer behavior, and product performance. You will write SQL queries to answer specific business questions and provide insights to management.

Learning Objectives

By completing this project, you will demonstrate proficiency in:

- Basic SQL queries (SELECT, WHERE, ORDER BY)
- Aggregate functions (COUNT, SUM, AVG, MIN, MAX)
- JOINs (INNER JOIN, LEFT JOIN, RIGHT JOIN)
- GROUP BY and HAVING clauses
- Subqueries and CTEs (Common Table Expressions)
- Window functions (RANK, ROW_NUMBER, PARTITION BY)
- Date functions and time-based analysis
- Business intelligence and data interpretation

2. Database Schema

Database Name: NovaRetailDB

Table 1: Products

Contains product catalog information (30 products).

Column	Data Type	Description
ProductID (PK)	VARCHAR(10)	Unique product identifier
ProductName	VARCHAR(100)	Product name
Category	VARCHAR(50)	Product category
UnitPrice	DECIMAL(10,2)	Selling price
CostPrice	DECIMAL(10,2)	Cost to acquire product

Table 2: Customers

Contains customer information (500 customers).

Column	Data Type	Description
CustomerID (PK)	VARCHAR(10)	Unique customer ID
FirstName, LastName	VARCHAR(50)	Customer name
Region	VARCHAR(20)	North, South, East, West
Channel	VARCHAR(20)	Online or Store
JoinDate	DATE	Date joined

Table 3: Sales

Contains transaction data (2,500 records). This is the main fact table.

Column	Data Type	Description
OrderID (PK)	VARCHAR(20)	Unique order ID
OrderDate	DATE	Transaction date
CustomerID (FK)	VARCHAR(10)	Links to Customers
ProductID (FK)	VARCHAR(10)	Links to Products

Quantity, UnitPrice	INT, DECIMAL	Order details
TotalSales, Profit	DECIMAL(10,2)	Financial metrics
Channel	VARCHAR(20)	Online or Store

Table 4: CustomerFeedback

Contains customer satisfaction surveys (1,000 records).

Column	Data Type	Description
FeedbackID (PK)	VARCHAR(20)	Unique feedback ID
OrderID, CustomerID	VARCHAR	Foreign keys
Rating	INT	1-5 star rating
Satisfaction	VARCHAR(50)	Satisfaction level
RecommendLikelihood	INT	1-10 NPS score

3. Setup Instructions

Create an ERD diagram here

Insert here <>

Step 1: Create Database(Ignore)

Run the SQL_Complete_Database_Setup.sql script to create all tables.

Step 2: Import Data

Import the CSV files for Customers, Sales, and CustomerFeedback tables using your database tool.

Step 3: Verify Import

Run the following query to verify all data is loaded:

```
SELECT 'Products' AS TableName, COUNT(*) AS Records FROM Products UNION ALL
SELECT 'Customers', COUNT(*) FROM Customers UNION ALL SELECT 'Sales',
COUNT(*) FROM Sales UNION ALL SELECT 'CustomerFeedback', COUNT(*) FROM
CustomerFeedback;
```

Expected Results:

- Products: 30
- Customers: 500
- Sales: 2,500
- CustomerFeedback: 1,000

4. Part 1: Basic SQL Queries (Beginner)

Write SQL queries to answer the following questions. Test each query and paste the results.

Question 1.1: Product Catalog (5 points)

List all products in the Electronics category. Show ProductID, ProductName, and UnitPrice. Order by price from highest to lowest.

Your SQL Query:

Question 1.2: Customer Count (5 points)

How many customers are in each region? Show Region and the count of customers.

Your SQL Query:

Question 1.3: Recent Orders (5 points)

Show the 10 most recent orders. Display OrderID, OrderDate, and TotalSales.

Your SQL Query:

Question 1.4: Affordable Products (5 points)

Find all products with a UnitPrice less than R1000. Show ProductName, Category, and UnitPrice.

Your SQL Query:

Question 1.5: Customer Satisfaction Summary (5 points)

Count how many feedback responses exist for each Satisfaction level. Order by count descending.

Your SQL Query:

5. Part 2: Intermediate SQL Queries

These questions require JOINs, aggregate functions, and GROUP BY clauses.

Question 2.1: Sales by Category (10 points)

Calculate total sales revenue for each product category. Show Category, Total Revenue, and Total Profit. Order by revenue descending.

Your SQL Query:

Question 2.2: Top Customers (10 points)

Find the top 5 customers by total purchase amount. Show CustomerID, Customer Name (FirstName + LastName), and Total Spent. Hint: Use CONCAT or + to combine names.

Your SQL Query:

Question 2.3: Monthly Sales Trend (10 points)

Show total sales by month for 2024. Display Year, Month (as number), and Total Sales. Order chronologically.

Your SQL Query:

Question 2.4: Channel Performance (10 points)

Compare Online vs Store performance. Show Channel, Number of Orders, Average Order Value, and Total Revenue.

Your SQL Query:

Question 2.5: Product Performance with Ratings (10 points)

For each product category, show the average customer rating. Include Category, Average Rating, and Number of Reviews. Only show categories with at least 50 reviews.

Your SQL Query:

6. Part 3: Advanced SQL Queries

These questions require subqueries, CTEs, window functions, or complex JOINs.

Question 3.1: Best Selling Product per Category (15 points)

Find the product with the highest total sales in each category. Show Category, ProductName, and Total Revenue for that product. Use a window function or subquery.

Your SQL Query:

Question 3.2: Customer Lifetime Value (15 points)

Create a query showing each customer's total purchases, number of orders, and average order value. Also include their region and primary channel. Only show customers with more than 3 orders. Order by total purchases descending.

Your SQL Query:

Question 3.3: Profit Margin Analysis (15 points)

Calculate the profit margin percentage for each product. Show ProductName, Category, Total Sales, Total Profit, and Profit Margin %. Order by profit margin descending.
Formula: (Profit / Sales) * 100

Your SQL Query:

Question 3.4: Year-over-Year Growth (15 points)

Compare sales between 2023 and 2024. Show the total sales for each year and calculate the growth percentage. Use a CTE or subquery.

Your SQL Query:

Question 3.5: Regional Performance Ranking (15 points)

Rank regions by total sales. Use the RANK() or ROW_NUMBER() window function. Show Region, Total Sales, Total Orders, and Rank.

Your SQL Query:

7. Part 4: Business Intelligence Questions

Answer these strategic questions using SQL queries and provide business insights.

Question 4.1: Customer Satisfaction vs. Repeat Purchases (20 points)

Analyze whether highly satisfied customers (rating 4-5) make more repeat purchases than less satisfied customers (rating 1-3). Show satisfaction level groups, average number of orders per customer, and total customers in each group.

Your SQL Query:

Business Insight (2-3 sentences):

Question 4.2: Discount Effectiveness (20 points)

Examine the relationship between discount percentage and profit. Create discount bands (0%, 1-10%, 11-20%, 21-30%) and show total sales, total profit, and profit margin for each band.

Your SQL Query:

Business Insight (2-3 sentences):

Question 4.3: Product Portfolio Optimization (20 points)

Identify underperforming products (bottom 5 by total revenue) and analyze whether they should be discontinued. Consider sales volume, profit margin, and customer ratings.

Your SQL Query:

Business Recommendation (3-4 sentences):

8. Part 5: Executive Management Report

Create a comprehensive report for Nova Retail Group's executive team. Your report should include SQL queries and written analysis.

Report Requirements (50 points total)

Section 1: Overall Business Performance (15 points)

Provide a summary of key business metrics:

- Total revenue and profit for 2023-2024
- Overall profit margin
- Total number of orders and customers
- Average order value

Your SQL Query:

Section 2: Top 3 Insights (20 points)

Identify and explain the three most important insights from your analysis. Each insight must be supported by SQL query results.

Insight #1:

Supporting SQL Query:

Insight #2:

Supporting SQL Query:

Insight #3:

Supporting SQL Query:

Section 3: Strategic Recommendations (15 points)

Based on your analysis, provide 3-5 actionable recommendations for management. Each recommendation should be data-driven.

9. Submission Guidelines

What to Submit

- This completed document with all SQL queries and answers
- A separate .sql file containing all your queries organized by section
- Screenshots of query results for Part 5 (Management Report)

Grading Criteria

Section	Points	Criteria
Part 1: Basic SQL	25	Correctness, syntax
Part 2: Intermediate SQL	50	JOINs, aggregations
Part 3: Advanced SQL	75	Subqueries, CTEs, windows
Part 4: Business Intelligence	60	Analysis, insights
Part 5: Management Report	50	Strategic thinking
TOTAL	**260**	

Tips for Success

- Test each query multiple times before submitting
- Use meaningful column aliases to make results clear
- Format your SQL code properly (indentation, line breaks)
- Add comments to complex queries explaining your logic
- Round decimal numbers appropriately (2 decimal places for currency)
- Think like a business analyst when interpreting results

Good Luck!